

# Uncertainty in AI: Basics of Probability

## Introduction

The journey of Artificial Intelligence has seen a significant evolution, transitioning from traditional, often deterministic, approaches to a more nuanced, modern paradigm that embraces uncertainty. Historically, AI systems relied on precise knowledge representation, where facts were unequivocally true or false, and states perfectly modeled the world. However, the real world is rarely so clear-cut. This lecture marks a pivotal shift towards understanding and integrating probabilistic reasoning into AI, acknowledging that perfect information is a luxury seldom afforded. By moving beyond rigid logical frameworks, modern AI gains the ability to reason with incomplete data, noisy observations, and inherent unpredictability, laying the groundwork for more robust and adaptable intelligent systems. This fundamental change allows AI to assign degrees of belief rather than absolute truths, enabling more sophisticated decision-making in complex, real-world scenarios.

## Core Concepts

| Term            | Definition                                                                                               | Significance                                                                                                                |
|-----------------|----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| State           | A complete description of the world or problem situation, often represented by assignments to variables. | Fundamental unit in traditional AI for navigating search spaces and modeling reality.                                       |
| Random Variable | A variable whose value is a numerical outcome of a random phenomenon.                                    | The basic unit of probability, capable of taking various values from a defined domain (e.g., heads/tails, 1-6, true/false). |
| Atomic Event    | A complete specification of the world, representing a full assignment of values to all random variables. | Analogous to a "state" in traditional AI, it's a mutually exclusive and exhaustive description of a possible world.         |

| Term                                  | Definition                                                                                                                  | Significance                                                                                                                        |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sample Space</b>                   | The set of all possible outcomes or atomic events for a given random variable or set of variables.                          | Defines the universe of possibilities within which probabilities are calculated.                                                    |
| <b>Prior Probability</b>              | The unconditional probability of an event, representing belief before any new evidence is observed.                         | Establishes an initial baseline belief about an event's likelihood without external information.                                    |
| <b>Joint Probability Distribution</b> | A probability distribution that gives the probability of all possible combinations of values for a set of random variables. | Provides a complete probabilistic model of the world, allowing derivation of any other probability query.                           |
| <b>Conditional Probability</b>        | The probability of an event occurring given that another event has already occurred. Also known as posterior probability.   | Crucial for updating beliefs based on new evidence, forming the basis of inference in uncertain environments.                       |
| <b>Independence</b>                   | Two events are independent if the occurrence of one does not affect the probability of the other.                           | Simplifies probabilistic models by indicating when certain pieces of evidence do not influence a variable of interest.              |
| <b>Chain Rule</b>                     | A fundamental rule for decomposing the joint probability of multiple variables into a product of conditional probabilities. | Essential for calculating joint probabilities and for building complex probabilistic models from simpler conditional relationships. |

## Detailed Analysis

### 1. The Imperative of Uncertainty in AI

The transition to modern AI is largely driven by the recognition that the world is inherently uncertain. While traditional AI often assumed a deterministic reality, where every outcome could be predicted given enough information, this perspective often falls short in practical applications.

- **Deterministic vs. Probabilistic Worldviews:** The lecture explores the philosophical debate about whether the world is fundamentally deterministic or probabilistic.
- A coin toss, for instance, might appear probabilistic (50% heads, 50% tails). However, a deterministic view suggests that if one could model all physical parameters (initial impulse, angle, air molecules, etc.), the outcome would be perfectly predictable.
- Conversely, quantum mechanics suggests inherent randomness at a fundamental level, implying a truly probabilistic universe.
- The practical implication is that regardless of the world's true nature, modeling all its complexities deterministically is often infeasible due to computational limits, data scarcity, and the sheer number of variables involved.
- **The Abstraction Dilemma:** AI systems must operate at a level of abstraction that is both manageable and useful.
- Attempting to model every minute detail of a system (e.g., every molecule in the air for a coin toss) is practically impossible.
- This leads to the famous adage: "All models are wrong, some are useful." Models are simplifications of reality, and while none perfectly capture every aspect, some provide sufficient accuracy for practical purposes.
- Probabilistic models offer a powerful way to handle the "less wrong" aspects by incorporating uncertainty and randomization for unmodeled complexities or unknown factors.
- **Limitations of Logic-Based AI:** Traditional logic-based systems, while powerful for well-defined problems, struggle with real-world uncertainty.
- Logic is "brittle"; it's difficult to incorporate exceptions to rules without making the system overly complex.
- It cannot easily integrate statistical properties or degrees of belief.
- Real-world scenarios involve noise in sensors, partial information, and unpredictable actions, which logic alone cannot effectively manage.

- **Data-Driven Necessity:** The rise of massive datasets in AI further necessitates probabilistic approaches.
- Learning from data often involves encountering unseen examples or variations, making deterministic rules insufficient.
- Probabilistic models allow AI to make statements about likelihoods rather than certainties, which is crucial when dealing with the inherent variability and incompleteness of real-world data.

## 2. Evolution of Knowledge Representation

The way AI represents knowledge has evolved significantly, moving from simple states to more complex logical and probabilistic frameworks.

- **States as Fundamental Units:** Early AI viewed the "state" as the core unit, modeling the world and problem situations.
- This allowed movement within a search space, but states were initially considered indivisible.
- **Exposing State Properties:** The need arose to expose properties within a state.
- **Constraint Satisfaction:** Introduced variables and values as parts of a state, allowing for more structured problem-solving.
- **Propositional Logic:** Modeled facts as true or false. If a fact's truth value was unknown, it was simply labeled "unknown."
- **First-Order Logic:** Extended propositional logic by introducing relations and objects, forming facts based on their interactions. Again, facts were strictly true, false, or unknown.
- **Beyond Binary Truths:** Recognizing the limitations of strict true/false assignments led to more flexible representations.
- **Temporal Logic:** Added the notion of time to facts, allowing for reasoning about sequences of events.
- **Fuzzy Logic:** Introduced the concept that facts themselves might not be absolutely true or false but could have degrees of truth.
- **The Probabilistic Paradigm:** This marked a significant shift, where facts, when unknown, are not just "unknown" but are assigned a "degree of belief" or probability.
- This allows for fine-grained reasoning and became the foundation of modern AI.

- Even first-order logic concepts can be combined with probability in "first-order probabilistic models" or "statistical relational learning," though these are advanced topics.

### 3. Fundamental Probability Definitions

Understanding basic probability concepts is essential for building probabilistic AI systems.

- **Random Variables:** These are the core units of probability, representing quantities whose values are outcomes of random phenomena. They can be discrete (e.g., sunny/rainy, 1-6) or continuous (e.g., temperature).
- **Atomic Events:** An atomic event is a complete specification of the world, meaning a full assignment of values to all random variables. These events are mutually exclusive (only one can occur at a time) and exhaustive (all possibilities are covered).
- **Sample Space (S):** This is the set of all possible atomic events or outcomes for a given random variable or set of variables. For example, rolling a die has a sample space of {1, 2, 3, 4, 5, 6}.
- **Probability Space/Model:** Defined by a sample space (S) and an assignment of probabilities  $P(x)$  for every  $x$  in S, such that:
  - $0 \leq P(x) \leq 1$  for all  $x$ .
  - The sum of  $P(x)$  over all  $x$  in S equals 1.
- **Event:** Any subset of the sample space. For example, "die roll less than 4" is an event, comprising outcomes {1, 2, 3}.
- **Basic Probability Formula:** For two events A and B, the probability of A or B occurring is given by:  

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$
 This accounts for the overlap (A and B) being counted twice.

### 4. Types of Probability Distributions

Probabilistic AI relies on different ways to express and calculate probabilities, depending on the available information.

- **Prior Probability (Unconditional Probability):** This is the probability of an event before any new evidence or information is considered. It reflects an initial belief.
- Example: "What is the probability Mr. Bean has a cavity?" or "What is the probability of sunny weather in September in Delhi?" These are initial beliefs without specific observations.

- A prior probability distribution assigns probabilities to all possible values of a single random variable (e.g.,  $P(\text{Weather} = \text{Sunny})$ ,  $P(\text{Weather} = \text{Cloudy})$ , etc., summing to 1).
- **Joint Probability Distribution:** This describes the probability of multiple random variables taking on specific values simultaneously. It represents the probability of an atomic event.
- Example: If modeling "Weather" and "Cavity," a joint distribution would specify  $P(\text{Weather}=\text{Sunny}, \text{Cavity}=\text{True})$ ,  $P(\text{Weather}=\text{Cloudy}, \text{Cavity}=\text{False})$ , etc.
- A joint distribution is incredibly powerful because, if known, it can answer any probabilistic query about the variables it covers.
- **Conditional Probability (Posterior Probability):** This is the probability of an event given that some evidence has already been observed. It updates the prior belief based on new information.
- Example: "What is the probability Mr. Bean has a cavity given he has a toothache?" The toothache is the new evidence.
- The formula for conditional probability is:  $P(A|B) = P(A \text{ and } B) / P(B)$ , where  $P(B) > 0$ . This means we are restricting our sample space to only those outcomes where B is true.
- Conditional probabilities are crucial for inference, allowing AI systems to adjust their beliefs as they gather more data.

## 5. Independence and the Chain Rule

Two critical concepts for simplifying and constructing probabilistic models are independence and the chain rule.

- **Independence:** Two random variables are independent if knowing the value of one provides no information about the value of the other.
- Example:  $P(\text{Cavity} | \text{Toothache}, \text{Weather}=\text{Sunny}) = P(\text{Cavity} | \text{Toothache})$  if "Weather=Sunny" is irrelevant to the probability of a cavity once a toothache is known.
- This concept is vital for simplifying models, as it means fewer conditional probabilities need to be specified or calculated.
- **Chain Rule (Product Rule):** This rule allows the decomposition of a joint probability distribution of multiple variables into a product of conditional probabilities.

- For variables  $X_1, X_2, \dots, X_n$ , the joint probability  $P(X_1, X_2, \dots, X_n)$  can be written as:  

$$P(X_1, \dots, X_n) = P(X_1) * P(X_2 | X_1) * P(X_3 | X_1, X_2) * \dots * P(X_n | X_1, \dots, X_{n-1})$$
- This rule is always true, regardless of the relationships between the variables. It's a fundamental identity that allows complex joint distributions to be expressed and computed from a series of conditional probabilities, which can often be easier to estimate or model.

## Key Takeaways

- [ ] Modern AI embraces uncertainty, moving beyond deterministic models to handle real-world complexities.
- [ ] Probabilistic models assign degrees of belief to unknown facts, enabling fine-grained reasoning.
- [ ] The choice of abstraction level is crucial; "all models are wrong, some are useful" highlights the necessity of simplification and probabilistic handling of unmodeled factors.
- [ ] Logic-based AI is brittle and struggles with exceptions and statistical properties, making probability a more robust foundation for uncertain environments.
- [ ] Random variables and atomic events are the fundamental building blocks for defining probabilistic spaces.
- [ ] Prior, joint, and conditional (posterior) probabilities are distinct but interconnected concepts for representing and updating beliefs.
- [ ] A joint probability distribution, if known, can answer any probabilistic query about the variables it covers.
- [ ] Independence simplifies models by identifying irrelevant evidence, while the chain rule is essential for decomposing and calculating joint probabilities.

## Conclusion

The shift to probabilistic AI is not merely an academic exercise but a practical necessity driven by the inherent uncertainty of the real world and the demands of data-intensive applications. From noisy sensor readings in robotics to incomplete patient data in medical diagnostics, AI systems constantly encounter situations where information is partial, observations are imperfect, and outcomes are not guaranteed. By adopting a probabilistic framework, AI gains the ability to quantify and reason about these uncertainties, making informed decisions even in the face of ambiguity. This foundation allows for the development of intelligent agents that can

learn from experience, adapt to changing conditions, and provide robust solutions in domains where traditional, deterministic approaches would fail. As AI continues to evolve, the principles of probability will remain central to building systems that are not only intelligent but also resilient and reliable in an unpredictable world.